Lecture 7

Tuesday, 31 January 2023 1:29 PM

Electrochemical Mechanisms

Similar process as heterogeneous catalysis → have to modify approach using echem kinetics

1) Series of elementary steps (1 e^- at a time)
2) Write rate law for each step
3) Assume RLS (rate limiting step), set all other rules to quasi-equilibrium
   (i) Eliminate intermediates / sites
4) Compose overall RL
5) Determine reversible potential (U), introduce η, & determine i_o
6) Compare i_o, α_a, α_c, and rxn order dependence to experiment

ex: Water Electrolysis → hydrogen evolution half rxn

Mechanism 1

Volmer: $S + H^+ + e^- \underset{k_{1,a}}{\overset{k_{1,c}}{\rightleftharpoons}} H_{ads}$

Tafel: $2H_{ads} \underset{k_{2,r}}{\overset{k_{2,f}}{\rightleftharpoons}} H_2(g) + 2S$

Mechanism 2

Volmer: $S + H^+ + e^- \rightleftharpoons H_{ads}$

Heyrovsky: $H_{ads} + H^+ + e^- \rightleftharpoons H_2(g) + S$

+ Tafel → non-faradaic
+ Both steps involve adsorbed species (i.e. surface sites) → [S]

Something we can measure (i.e. BET isotherm)

Volmer-Tafel Mechanism:

$$r_1 = k_{1,c}[H^+][S]\exp\left[\frac{-\beta FV}{RT}\right] - k_{1,a}[H_{ads}]\exp\left[\frac{(1-\beta)FV}{RT}\right]$$

C_s = # unoccupied sites $= C_t(1-\theta)$; $C_{H_{ads}} = C_t\cdot\theta$

fractional coverage $\left[\frac{C_{H_{ads}}}{C_t}\right]$

$$\rightarrow r_1 = k_{1,c}C_{H^+}C_t(1-\theta)\exp\left[\frac{-\beta FV}{RT}\right] - k_{1,a}C_t\theta\exp\left[\frac{(1-\beta)FV}{RT}\right]$$

$$r_2 = k_{2,f}[H_{ads}]^2 - k_{2,r}[H_2][S]^2$$

$$= k_{2,f}C_t^2\theta^2 - k_{2,r}C_{H_2}[C_t(1-\theta)]^2$$

$$\rightarrow r_2 = k_{2,f}(C_t\theta)^2 - k_{2,r}P_{H_2}[C_t(1-\theta)]^2$$

Solving for θ:

r_2 is rate limiting, then r_1 is in quasi-equil.: $r_1 = 0 \Rightarrow k_{1,c}C_{H^+}C_t(1-\theta)\exp\left[\frac{-\beta FV}{RT}\right] = k_{1,a}C_t\theta\exp\left[\frac{(1-\beta)FV}{RT}\right]$

$$\theta = \frac{k_{1,c}C_{H^+}}{k_{1,a}\exp\left[\frac{FV}{RT}\right] + k_{1c}C_{H^+}}$$

$$\theta = \frac{\frac{k_{1,c}C_{H^+}}{k_a}\exp\left[\frac{-FV}{RT}\right]}{1 + \frac{k_{1c}C_{H^+}}{k_a}\exp\left[\frac{-FV}{RT}\right]}$$

→ similar to Langmuir isotherm $\theta = \frac{K}{1+K}$

Net rate:

$$r_{net} = r_2 = \frac{i}{2F} = k_{2,f}(C_t\theta)^2 - k_{2,r}P_{H_2}[C_t(1-\theta)]^2$$

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Even though has no e^-, it is still used to define r_{net} and we use this Faraday's eq. always.

$\theta = f(V)$ [depends on applied potential] !

↳ Non-faradaic processes can also be indirectly affected by potential ★

This expression is not consistent w/ B-V kinetics

↳ There might be deviations from B-V (& rare exceptions)

Could be other

- Volmer (rds) - Tafel
- Volmer (rds) - Heyrovsky
- Volmer - Heyrovsky (rds)

} These all have e^- transfer on rds ↓ will take the look of B-V !

Ex: $Cu \rightarrow Cu^{2+} + 2e^-$

1. $Cu \underset{k_c}{\overset{k_{1a}}{\rightleftharpoons}} Cu^+ + e^-$

2. $Cu^{2+} \underset{k_{2c}}{\overset{k_{2a}}{\rightleftharpoons}} Cu^{2+} + e^-$ [RDS] ; $\beta = 0.5 \rightarrow \alpha_a = 1.5 \rightarrow i_0 \sim C_{Cu^{2+}}^{3/4}$ ← Show this

$\alpha_c = 0.5$